



Instant3D: A Lightweight Open-Source GUI Frontend for TotalSegmentator-Based Medical Image Segmentation and 3D Reconstruction

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Abstract

Automatic segmentation is increasingly used in medical imaging, but many tools require command-line operation and environment setup. TotalSegmentator provides open-source multiorgan segmentation for computed tomography (CT) and magnetic resonance imaging (MRI) data, yet its routine use may be challenging for nontechnical users. Herein, we describe the design, implementation, and functionality of Instant3D, a lightweight, open-source graphical user interface (GUI) frontend for TotalSegmentator. Instant3D was developed in Python using PyQt6. It accepts DICOM folders, NIfTI files, and NRRD files as input and allows users to select regions of interest (ROIs) through a suggestion-enabled interface. The GUI executes TotalSegmentator as the backend. Default outputs include STL meshes for 3D visualization and NIfTI segmentation images, with optional SVG masks and CSV files containing volumetric measurements. SVG masks are interoperable with SegRef3D for slice-level review and refinement of automated segmentation results. Batch processing is supported. Functionality was tested using representative CT and MRI datasets. Instant3D successfully imported supported formats, executed TotalSegmentator through the GUI, and generated expected outputs, including STL, NIfTI, SVG, and CSV files. Representative datasets demonstrated successful workflow completion and compatibility with SegRef3D. Batch processing generated organized output files for multiple input datasets, confirming workflow feasibility. Instant3D simplifies TotalSegmentator execution and facilitates generation of reusable 3D reconstruction outputs. While not a replacement for comprehensive visualization platforms, it complements them by providing a streamlined workflow for ROI selection, batch processing, and multifformat export. Instant3D may serve as a practical tool for research, education, quantitative imaging, and 3D model generation.

Keywords Automatic segmentation · 3D reconstruction · Medical imaging · Graphical user interface (GUI) · Radiomics · 3D printing

Background

Automatic segmentation is increasingly important in medical imaging, as it supports quantitative image analysis, organ volumetry, radiomics, surgical simulation, educational visualization, and the generation of patient-specific three-dimensional (3D) models [1–10]. Recent deep learning-based frameworks have markedly improved the feasibility of multiorgan segmentation [11–13]. Among them, TotalSegmentator has emerged as a practical open-source tool for automated segmentation of major anatomical structures in computed tomography (CT) and magnetic resonance imaging (MRI) datasets [14–16]. The original CT implementation was based on the nnU-Net framework and trained on a large, clinically heterogeneous set of CT examinations, enabling robust segmentation across a range of anatomical

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structures [11, 15]. More recently, TotalSegmentator MRI has extended this approach to sequence-independent MRI segmentation [16]. These features make TotalSegmentator well suited for clinical research, education, and 3D reconstruction workflows. However, its routine use often requires command-line operation, environment setup, and familiarity with model options, which may limit adoption among clinicians, educators, and students without programming experience.

TotalSegmentator is one of several approaches to automated or artificial intelligence–assisted medical image segmentation. General frameworks such as MONAI and MONAI Label provide flexible infrastructure for model development, deployment, and interactive labeling [12], while foundation model-based approaches, including Segment Anything Model (SAM)–related methods, have generated significant interest in promptable segmentation [17]. In addition, recent studies have proposed advanced model architectures and training strategies for medical image analysis and segmentation, including diffusion-based translation models [18], modality-aware transformers [19], hybrid masked image modeling [20], and Mamba-based long-range sequential modeling [21]. Although these frameworks are powerful and extensible, their practical use often requires model selection, configuration, scripting, or adaptation to specific anatomical targets and imaging modalities. In contrast, TotalSegmentator provides readily available pretrained models with broad anatomical coverage, making it particularly suitable as the backend for a lightweight graphical user interface (GUI) designed for immediate multiorgan segmentation and 3D output generation.

Graphical platforms such as 3D Slicer and OsiriX/OsiriX MD are well-established solutions for image visualization and segmentation. 3D Slicer provides a comprehensive suite of manual, semiautomatic, and artificial intelligence–assisted tools, and TotalSegmentator can be integrated within this environment through a dedicated extension [15, 16]. Other interactive or web-based segmentation and visualization tools, including ITK-SNAP, Seg3D, Biomedisa, and OHIF, also play important roles in medical image analysis workflows [22–25]. These platforms are highly capable of visualization, manual correction, and comprehensive image analysis; however, they are not specifically optimized for a minimal workflow centered on rapid TotalSegmentator execution, batch processing, and direct export of 3D reconstruction–oriented output files. OsiriX/OsiriX MD also offers plugin-based segmentation functionality, but its clinical-grade use is limited by platform and licensing constraints. Therefore, there remains a practical need for a lightweight GUI frontend that complements, rather than replaces, existing platforms.

To address this need, we developed Instant3D, an open-source GUI designed as a lightweight frontend for

TotalSegmentator. Instant3D emphasizes intuitive region of interest (ROI) selection, batch processing, and direct multiformat export. STL files provide surface meshes suitable for 3D visualization, 3D printing, surgical simulation, and downstream modeling [1, 6–10]. SVG masks provide editable, slice-level segmentation data that can be reviewed and manually refined in SegRef3D [26, 27]. CSV files provide volumetric measurements for quantitative analysis and dataset-level comparison. In this way, Instant3D aims to bridge automated segmentation and practical 3D reconstruction workflows by reducing the number of manual steps required to generate reusable outputs. The aim of this Technical Note was to describe the design, implementation, representative functionality testing, and intended use of Instant3D.

Methods

Software Architecture and Distribution

Instant3D was developed in Python (version 3.12) [28] using the PyQt6 framework [29] for the GUI. Core libraries include nibabel [30] and pydicom [31] for image handling, SimpleITK [32] for preprocessing, and trimesh [33] for mesh processing. The application is packaged as a standalone Windows executable using PyInstaller [34] and distributed through GitHub (<https://github.com/SatoruMuro/Instant3D>), allowing users to run the GUI without setting up an Instant3D-specific Python environment.

TotalSegmentator is used as the segmentation backend and must be installed separately. At present, this initial setup requires command-line operations, including the installation of TotalSegmentator and downloading the required model weights. Once configured, Instant3D automatically invokes the TotalSegmentator backend through the GUI, thereby reducing the need for command-line interaction during routine use. The current distribution primarily targets Windows users; broader packaging, simplified installation, Docker-based distribution, and more comprehensive backend bundling are planned as future development goals.

Core Workflow

Instant3D is designed around a simple three-step workflow: data import, ROI selection, and execution (Fig. 1). For data import, the software accepts DICOM folders, NIfTI files, and NRRD files. When necessary, input data are automatically converted to the NIfTI format for processing.

ROI selection is facilitated by a suggestion-enabled input interface with alias recognition. During text entry, the ROI field suggests valid TotalSegmentator ROI names, while also allowing users to search using commonly used aliases, partial terms, or related anatomical keywords. This

Quick 3 Steps on GUI

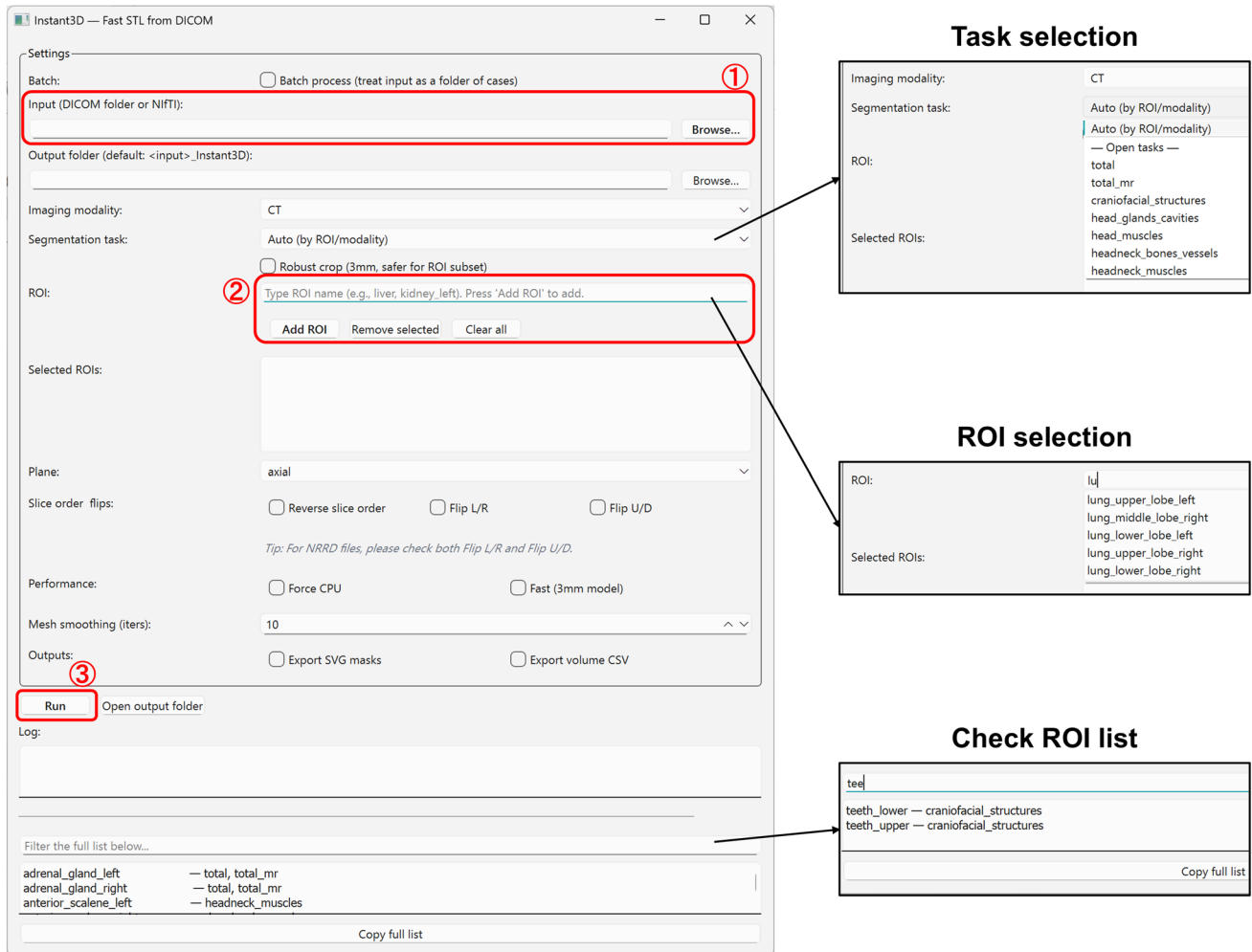


Fig. 1 Instant3D graphical user interface. The ROI input field, Run button, and optional settings panel are shown, illustrating the three-step workflow for data import, ROI selection, and execution. GUI, graphical user interface; ROI, region of interest

design helps nonexpert users identify the intended anatomical structure without requiring knowledge of exact backend label names. Multiple ROIs can be added sequentially using the “Add ROI” function.

During execution, a single button press runs TotalSegmentator through the GUI. By default, Instant3D generates STL meshes and segmentation images in the NIfTI format. When the corresponding options are enabled, it can additionally export per-slice SVG segmentation masks and CSV files containing volumetric measurements. Output files are named consistently and organized into structured output directories. When multiple ROIs are selected, each ROI is processed and exported as a separate output, and the corresponding volumetric measurements are summarized by ROI in the CSV file.

Advanced Options

Instant3D includes several advanced options to support both simple and more flexible workflows. Batch processing enables users to sequentially process multiple input datasets without writing independent scripts or manually running each case. The imaging modality setting allows users to switch between CT and MRI processing modes according to the input data and the selected TotalSegmentator backend.

The segmentation task option allows users to select among available TotalSegmentator subtasks, such as total, total_mr, craniofacial structure-related tasks, and head muscle-related tasks, depending on the installed TotalSegmentator version and supported backend models. Multiple ROI selection enables several anatomical structures to be

processed within a single workflow. Plane specification allows users to select axial, coronal, or sagittal orientations when generating slice-based outputs. Slice order and flip options enable users to reverse slice order or flip images left–right or up–down as needed for compatibility with downstream visualization or editing workflows.

Performance settings include options for central processing unit (CPU) execution and Fast processing. The CPU option forces processing on the CPU when a compatible graphics processing unit is unavailable, although this generally increases the processing time. The Fast option uses the lower-resolution 3-mm TotalSegmentator model to reduce processing time. Mesh smoothing can be adjusted by specifying the number of smoothing iterations, with a default value of 10.

Output options allow users to export additional file formats. Per-slice SVG masks can be exported for compatibility with SegRef3D, and volumetric CSV files can be generated for quantitative analysis. The volumetric CSV file is a table listing the selected ROI names and their corresponding volume measurements, enabling dataset-level comparison and downstream statistical analysis. A built-in ROI/task search utility is also provided to help users confirm supported ROI names and available tasks within the GUI.

Interoperability with SegRef3D

Instant3D is designed to interoperate with SegRef3D, an open-source platform for artificial intelligence–assisted segmentation, interactive refinement, and 3D reconstruction in morphological research [26, 27]. In this workflow, Instant3D exports per-slice SVG masks derived from automated segmentation results. These SVG masks can then be opened in SegRef3D for slice-level visualization and manual editing. After refinement, SegRef3D can be used to generate revised 3D reconstruction outputs. This interoperability allows users to combine automated segmentation with user-driven correction when segmentation boundaries require anatomical review or refinement.

Functionality Testing

Because this Technical Note focuses on software design and workflow demonstration, Instant3D was evaluated through functionality testing rather than formal segmentation accuracy validation. The purpose of this testing was to confirm whether the GUI could successfully import medical image data, execute TotalSegmentator, generate the expected output files, and perform batch processing. The test categories and confirmation criteria are summarized in Table 1.

Instant3D was tested using representative CT and MRI datasets, including publicly available example data from the 3D Slicer Sample Data module (<https://www.slicer.org>). The tested inputs included DICOM folders and pre-converted NIfTI or NRRD files. Representative anatomical targets included abdominal organs, thoracic organs, the skull and teeth, and brain structures. For each test, we confirmed successful data import, ROI selection, segmentation execution, STL mesh generation, NIfTI segmentation image output, optional SVG mask export, and optional CSV volume export. Batch functionality was evaluated by sequentially processing multiple input datasets and confirming completion of the expected outputs for each dataset.

This functionality testing was intended to evaluate software behavior and workflow operation. It was not designed to assess segmentation accuracy against expert manual labels or to compare Dice coefficients, Hausdorff distances, or other segmentation performance metrics. The segmentation accuracy of the backend model depends on TotalSegmentator and has been evaluated in previous studies [15, 16].

Results and Demonstration

Generated Output Files

Instant3D generated the expected output files from representative CT and MRI datasets. By default, the software generated STL meshes and segmentation images in the NIfTI format for the selected ROIs. When additional output

Table 1 Summary of functionality testing

Test category	Input/target	Confirmed functionality
CT dataset	Abdominal organs	Data import, ROI selection, TotalSegmentator execution, STL/NIfTI output, optional SVG/CSV output
CT dataset	Thoracic organs	Data import, ROI selection, TotalSegmentator execution, STL/NIfTI output, optional SVG/CSV output
CT dataset	Skull and teeth	Data import, ROI selection, TotalSegmentator execution, STL/NIfTI output
MRI dataset	Brain structures	Data import, ROI selection, TotalSegmentator execution, STL/NIfTI output, optional SVG/CSV output
Batch processing	Multiple input datasets	Sequential processing and generation of output directories and files for each dataset

CT computed tomography, ROI region of interest

options were enabled, Instant3D also generated per-slice SVG segmentation masks and CSV files containing volumetric measurements.

The STL files provided surface mesh models of the selected anatomical structures and could be opened in external 3D visualization software such as 3D Slicer (Fig. 2). Representative outputs included abdominal organs, brain structures, the skull and teeth, and thoracic organs. The NIfTI segmentation images provided voxel-based label data corresponding to the automated segmentation results. The optional SVG masks provided slice-level, editable segmentation contours and were confirmed to be compatible with SegRef3D for manual review and refinement (Fig. 3). The optional CSV files listed the selected ROI names and their corresponding volume measurements, allowing quantitative comparison across datasets.

Functionality Testing Results

All functionality checks were successfully completed (Table 1). Batch processing generated a separate, correctly organized output directory and the expected files for each input dataset.

Workflow Characteristics and Relationship with Existing Platforms

The default workflow allows users to generate segmentation outputs through three steps: data import, ROI selection, and execution via the Run button. Optional advanced settings, including modality selection, TotalSegmentator task

selection, multiple ROI selection, batch processing, performance settings, and output toggles, provide additional flexibility within the same GUI.

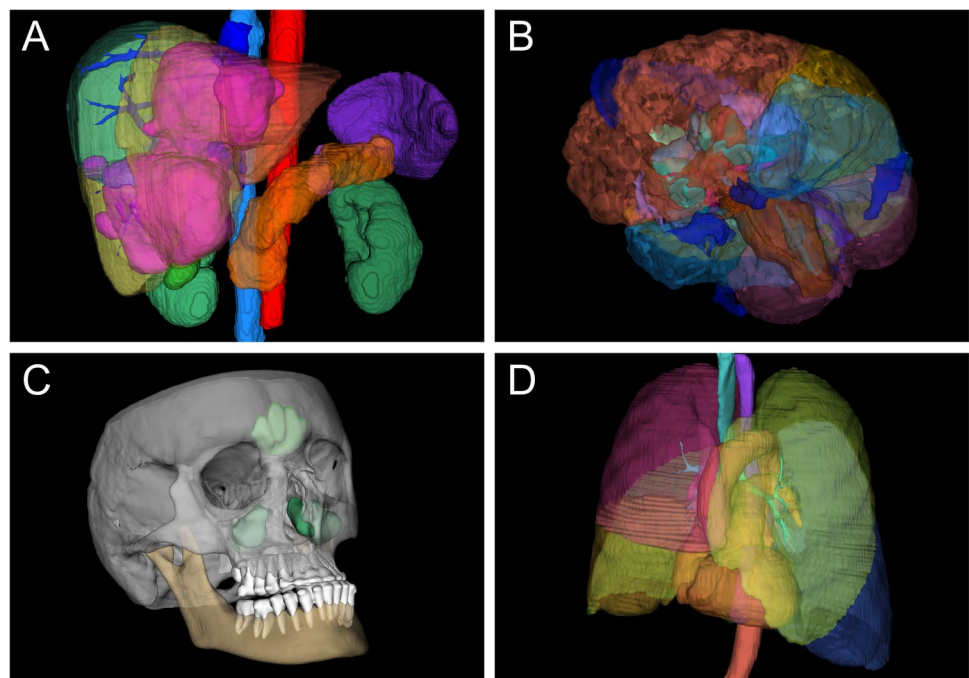
Instant3D is not intended to replace comprehensive visualization and segmentation platforms. Rather, it is designed as a lightweight frontend for users requiring rapid TotalSegmentator execution, batch processing, and direct generation of reusable output files. In this workflow, external platforms remain important for downstream visualization, manual refinement, and more advanced image analysis. The relationship between Instant3D and related platforms is summarized in Table 2.

Discussion

Instant3D provides an open-source, user-friendly GUI frontend that connects TotalSegmentator-based automatic segmentation with practical 3D reconstruction workflows in research and education. By simplifying data import, ROI selection, backend execution, and output generation through a streamlined workflow, the tool reduces the need for command-line operation during routine use. In addition to default STL mesh and NIfTI segmentation image outputs, optional SVG masks and CSV volume data extend its functionality and support interoperability with SegRef3D for interactive refinement.

Comprehensive platforms such as 3D Slicer, OsiriX/ OsiriX MD, ITK-SNAP, Seg3D, Biomedisa, and OHIF play important roles in medical image visualization, segmentation, manual correction, and clinical image review

Fig. 2 Three-dimensional STL models generated by Instant3D. Representative STL models of abdominal organs (A), brain structures (B), the skull and teeth (C), and thoracic organs (D) generated by Instant3D. Instant3D itself does not include an STL viewer; the models were visualized using 3D Slicer



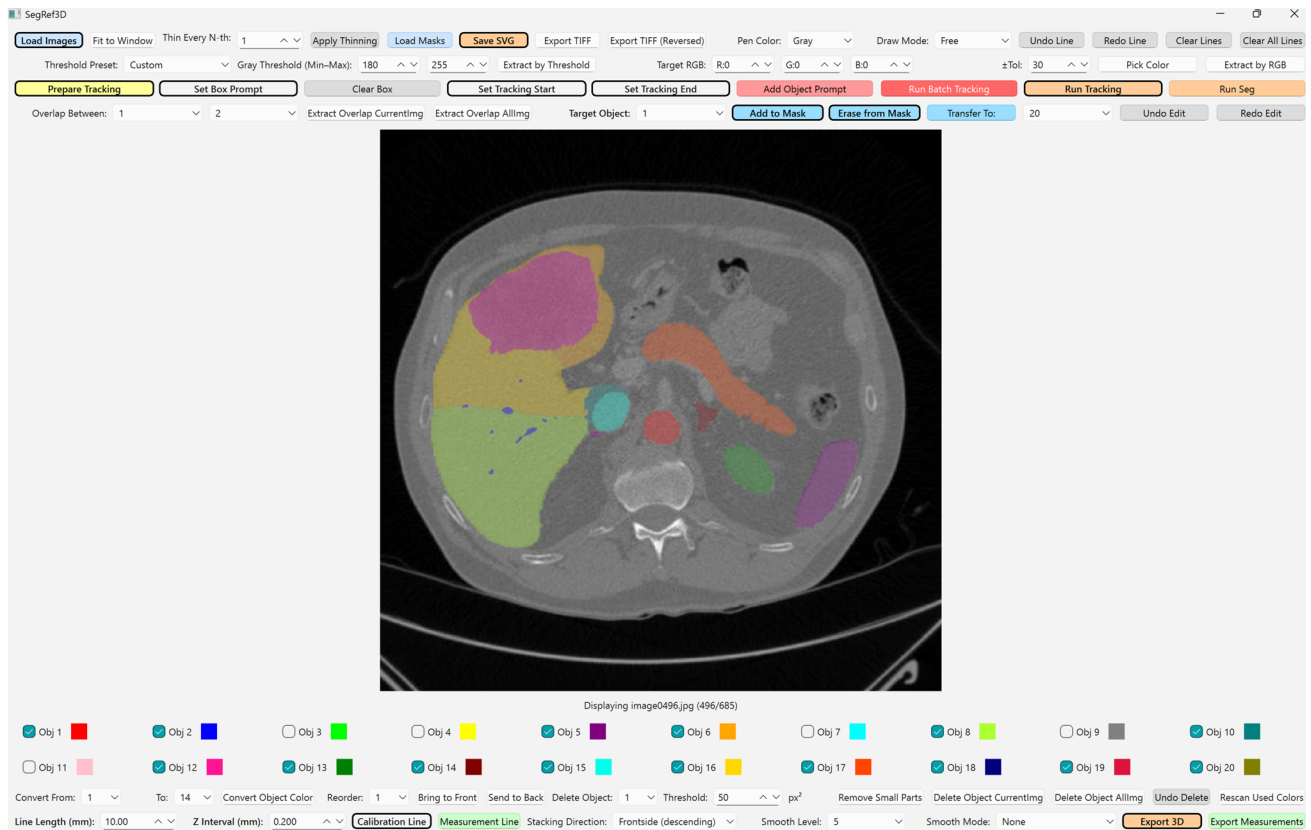


Fig. 3 Compatibility with SegRef3D for interactive refinement. Representative slice image with an overlaid SVG mask exported from Instant3D. The SVG mask can be opened in SegRef3D for slice-by-

slice visualization, manual editing, and refinement of the automated segmentation results. After mask refinement, SegRef3D can export the updated data as STL models

Table 2 Comparison of Instant3D with related software platforms

Feature	Instant3D	3D Slicer+ TotalSegmentator	OsiriX/OsiriX MD	ITK-SNAP/Seg3D/Biomedisa/OHIF
Primary role	Lightweight TotalSegmentator frontend and output generator	Comprehensive image analysis and visualization platform	DICOM viewing and plugin-based image analysis	Interactive segmentation, visualization, or web-based image review
TotalSegmentator execution	Supported through GUI	Supported through extension	Plugin-based support	Not the primary workflow
Batch processing	Built-in workflow	Possible but less streamlined for nonexpert users	Limited or workflow-dependent	Limited or tool-dependent
Direct STL output	Supported	Possible through export workflows	Workflow-dependent	Tool-dependent
NIfTI segmentation output	Supported	Supported	Workflow-dependent	Tool-dependent
SVG mask output	Supported for SegRef3D-compatible refinement	Not a primary output	Not a primary output	Not a primary output
CSV volume output	Supported	Possible through additional tools or modules	Workflow-dependent	Tool-dependent
Interactive visualization	Not included	Strong	Strong	Strong or tool-dependent
Best suited use case	Rapid segmentation execution, batch output, and reusable 3D reconstruction files	Detailed visualization, manual correction, and comprehensive image analysis	Clinical DICOM viewing and macOS-based workflows	Interactive segmentation, annotation, and web-based image review

GUI graphical user interface, 3D three dimensional

[22]. These tools are not replaced by Instant3D; rather, Instant3D is intended to complement them by focusing on a narrower workflow: rapid TotalSegmentator execution, batch processing, and direct export of reusable STL, NIfTI, SVG, and CSV outputs. This distinction is important because comprehensive platforms, while powerful, may require multiple modules, manual export steps, or familiarity with complex image-analysis environments. Instant3D therefore occupies a practical niche as a lightweight, output-oriented frontend for users who require rapid generation of 3D reconstruction files rather than a full visualization or image-analysis platform.

One potential application of Instant3D is in quantitative imaging and radiomics research, where reproducible segmentation and scalable extraction of volumetric or shape-related features are important preprocessing steps [2–5]. Instant3D may streamline this process by enabling users to run TotalSegmentator through a GUI and export STL meshes, NIfTI segmentation images, and CSV-based volumetric measurements. In addition, SVG masks can be reviewed and refined in SegRef3D, supporting quality control when automated segmentation results require anatomical inspection [26, 27]. However, Instant3D itself does not introduce a new segmentation model, and the accuracy of the segmentation results depends on the TotalSegmentator backend.

The combination of batch processing and multiformat export may be particularly useful for studies that require repeated dataset processing or downstream applications such as 3D printing and extended-reality prototyping [6–10].

This study has certain limitations. First, as this manuscript is presented as a Technical Note, the evaluation was limited to functionality testing using representative datasets. Formal segmentation accuracy metrics, such as Dice coefficients or Hausdorff distances against expert manual labels, were not assessed. Second, formal user-performance metrics, usability testing, and time-based comparisons with other platforms were not performed. Third, Instant3D functions as a frontend for TotalSegmentator; therefore, the range of segmentable structures and overall segmentation accuracy are constrained by the installed backend. Fourth, the current packaged distribution primarily targets Windows users, and initial backend installation still requires some command-line operations. Fifth, Instant3D does not include an integrated 3D viewer; external software is required for detailed visualization and interactive analysis. Finally, several features remain as future development goals, including detailed CPU usage monitoring, broader cross-platform packaging, Docker-based distribution, more comprehensive backend bundling, multibackend support for comparing different segmentation methods, tumor-specific segmentation workflows, and regulatory or clinical validation.

Conclusions

We developed Instant3D, an open-source and user-friendly GUI frontend for TotalSegmentator-based medical image segmentation and 3D reconstruction. Instant3D provides a streamlined workflow for data import, ROI selection, backend execution, batch processing, and direct export of STL meshes, NIfTI segmentation images, SVG masks, and CSV volume tables. Although it does not replace comprehensive visualization or segmentation platforms, it complements them by simplifying the generation of reusable, 3D reconstruction-oriented outputs. Instant3D may therefore serve as a practical workflow tool for research, education, quantitative imaging, 3D printing, and prototype development for virtual or mixed reality applications.

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Data Availability The source code used in this study is openly available on GitHub at <https://github.com/SatoruMuro/Instant3D>. The repository contains the full implementation of the algorithms, graphical user interface, and processing workflows required to reproduce the analyses described in this paper. All code is released under an open license and can be accessed without restriction. No proprietary data or materials were used in this study.

Declarations

Competing interests The authors declare no competing interests.

Usage of Generative AI and AI-Assisted Technologies During the preparation of this manuscript, ChatGPT was used to improve the clarity and grammatical quality of the English language. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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